



Find the state of the art on how prompt interventions affect Multimodal LLM / Vision-Language Model behavior, with emphasis on both:

- (1) output quality/performance, and
- (2) explainability, grounding, attribution, saliency, activation maps, and explanation stability.

My project is about stress-testing explainability in MLLMs. The specific subtopic is prompt ablation: how wording, structure, ordering, length, decomposition, chain-of-thought, role prompting, self-refinement, and other prompt interventions affect multimodal grounding and TAM-style token-level visual explanations.

Prioritize:

- recent MLLM/VLM/IVLM work,
- Qwen2-VL, LLaVA, InternVL when possible,
- papers directly on prompting + multimodal explainability,
- but also include broader methodology papers, surveys, literature reviews, benchmarks, datasets, code, and supplementary material when relevant.

Search broadly for:

- prompt sensitivity
- prompt robustness
- prompt brittleness
- prompt optimization
- multimodal prompting
- visual grounding
- saliency / attribution / activation maps
- causal / probing / hidden-state interpretability
- multimodal explainability
- faithfulness vs plausibility
- failure cases / bias / instability

For each result, label it as:

- directly relevant: MLLM + prompt + explainability
- directly relevant: MLLM + prompt + performance
- relevant methodology
- survey/review
- benchmark/dataset/code/supplementary
- non-peer-reviewed useful source

For each item, give title, year, venue/source, short contribution summary, why it matters for this project, and whether it informs output quality, explainability/grounding, or both.

Then synthesize:

- main thematic clusters
- most relevant papers
- best surveys/reviews
- transferable methodology papers
- useful benchmarks/code

- open gaps for prompt ablation studies on TAM-like explainability in MLLMs

Here is a state-of-the-art map of **prompt interventions in MLLMs/VLMs**, with emphasis on output quality and on explainability/grounding/attribution stability. The strongest takeaway is that prompt wording and structure can materially change both answer quality and the internal/external explanations you get, but most current work studies these effects indirectly; direct, systematic prompt-ablation studies for TAM-like **token-level visual explanations** are still scarce.^{[1] [2] [3]}

Thematic clusters

1) Visual prompting and prompt tuning

This cluster covers learned or engineered visual prompts, prompt tuning, and prompt placement. The most relevant recent survey is *Visual Prompting in Multimodal Large Language Models: A Survey* (2024), which organizes visual prompting, prompt generation, compositional reasoning, and prompt learning, while explicitly linking prompting to grounding and object referring. A closely related methodology line is *Rethinking Visual Prompting for Multimodal Large Language Models* (2024), which ablates different prompt insertion strategies and shows that prompt design changes performance and stability.^{[4] [5]}

2) Prompt sensitivity and brittleness

A smaller but important body of work tests whether model behavior changes under prompt variations such as order, phrasing, or decomposition. *Understanding the Multi-modal Prompts of the Pre-trained Vision-Language Model* (2023/2024) probes multi-modal prompts with attention, attention distance, entropy, and visualization, and concludes that prompts often capture dataset bias rather than truly robust semantics. That is highly relevant for prompt ablation because it suggests that some prompt gains may be superficial and unstable under paraphrase or structure changes.^[6]

3) Grounding and localization

This cluster focuses on prompt-based visual grounding, object referring, and coordinate prediction. Qwen2-VL and Qwen2.5-VL explicitly emphasize precise object grounding and structured coordinates/JSON-style outputs, making them especially relevant for prompt studies where the prompt asks the model to localize evidence rather than just answer textually. Work like *Visual Position Prompt for MLLM based Visual Grounding* (2025) adds spatial cues to improve grounding, showing that prompt-like spatial intervention can improve localization quality.^{[6] [7] [8]}

4) Explainability and attribution

This is the most directly relevant cluster for your project. *Explainable and Interpretable Multimodal Large Language Models: A Comprehensive Survey* (2024) is a strong anchor because it organizes interpretability work across data, model, and training/inference perspectives, and covers token-level to embedding-level interpretation. *Token Activation Map (TAM) to Visually Explain Multimodal LLMs* (2025) is especially relevant because it is explicitly token-level, multimodal, and designed for multiple generation rounds and failure analysis. A complementary method paper, *Gradient-Layer Importance Mapping for Prompted Visual Saliency Explanation* (2025), targets open-ended VQA and jointly attributes outputs to visual and textual signals.^{[2] [9] [3] [10]}

5) Benchmarks and evaluation

For explainability, the field still lacks a standard benchmark that measures not just saliency quality, but stability under prompt changes. *Saliency-Bench* provides standardized evaluation for saliency methods using ground-truth explanations and faithfulness/alignment metrics such as IoU, Pointing Game, insertion AUC, and reliability analysis. For multimodal causal or grounded reasoning, newer benchmarks like *ViLCaR* (2026) point toward graph-aligned causal attribution and inference evaluation, which is conceptually very close to prompt-ablation stress tests even though it is not specifically about prompts.^{[11] [12] [13]}

Most relevant papers

Item	Label	Year	Venue/source	Contribution	Why it matters	Informs
<i>Token Activation Map to Visually Explain Multimodal LLMs</i>	directly relevant: MLLM + prompt + explainability	2025	arXiv / ICCV 2025 Oral	Token-level visual explanation method for MLLMs; supports multiple generation rounds, object localization, and failure analysis [3] [10] .	This is one of the closest matches to TAM-style visual explanations and is central for studying explanation stability under prompt changes.	Both
<i>Gradient-Layer Importance Mapping for Prompted Visual Saliency Explanation</i>	directly relevant: MLLM + prompt + explainability	2025	arXiv	Jointly attributes open-ended VQA outputs to visual and textual signals; designed for prompted saliency explanation [9] .	Useful for measuring whether prompt wording changes what evidence the model appears to use.	Both
<i>Understanding the Multi-modal Prompts of the Pre-trained Vision-Language Model</i>	directly relevant: MLLM + prompt + performance	2023/2024	arXiv	Probes multimodal prompt mechanics with attention, entropy, and visualization; finds prompt learning often encodes dataset bias [5] .	Strong for prompt brittleness and for designing ablation protocols around prompt semantics versus dataset artifacts.	Both
<i>Rethinking Visual Prompting for Multimodal Large Language Models</i>	directly relevant: MLLM + prompt + performance	2024	arXiv	Compares several prompting strategies and prompt insertion variants; reports consistent gains from their visual prompt design [4] .	Useful for structured ablations over prompt placement and modality of intervention.	Both
<i>Visual Position Prompt for MLLM based Visual Grounding</i>	directly relevant: MLLM + prompt + performance	2025	arXiv / TMM	Adds global and local position prompts to improve grounding and coordinate prediction [8] .	Good example of prompt-like spatial intervention that can be evaluated with localization accuracy and explanation maps.	Both
Qwen2-VL / Qwen2.5-VL model papers	directly relevant: MLLM + prompt + performance	2024–2025	arXiv / Qwen blog	Emphasize perception, grounding, and structured coordinate outputs [6] [7] .	Useful baseline family for prompt ablations because grounding is a first-class capability.	Both

Item	Label	Year	Venue/source	Contribution	Why it matters	Informs
<i>Explainable and Interpretable Multimodal Large Language Models: A Comprehensive Survey</i>	survey/review	2024	arXiv	Broad taxonomy of interpretability and explainability in MLLMs across data/model/training-inference [2] .	Best single overview for framing prompt-ablation explainability questions.	Both
<i>Visual Prompting in Multimodal Large Language Models: A Survey</i>	survey/review	2024	arXiv	Survey of visual prompting methods, compositional reasoning, and grounding/object referring [1] .	Best prompt-centric survey for MLLMs.	Performance + grounding
<i>Saliency-Bench</i>	benchmark/dataset/code/supplementary	2023	arXiv	Benchmark for saliency methods with alignment and faithfulness metrics [12] [13] .	Transferable evaluation framework for explanation quality and reliability under prompt perturbations.	Explainability/gr
<i>ViLCaR</i>	benchmark/dataset/code/supplementary	2026	arXiv	Diagnostic benchmark for causal attribution, inference, and QA in LVLMs [11] .	Highly relevant conceptually for causal attribution under prompt interventions.	Explainability/gr

Best surveys

The two surveys worth reading first are *Explainable and Interpretable Multimodal Large Language Models* and *Visual Prompting in Multimodal Large Language Models*. The first is better for the explainability stack, including token/embedding analysis and training/inference interventions; the second is better for prompt engineering and visual prompting mechanisms. Together, they frame both the input-side and internal-mechanism view of prompt interventions. [\[1\]](#) [\[2\]](#)

Transferable methodology

For your project, the most transferable methods are attention/entropy analyses, perturbation tests, and attribution methods that jointly examine text and image evidence. TAM is especially useful because it is designed to show token-by-token visual contributions across generation rounds, which makes it a natural target for prompt stability studies. Saliency-Bench is also valuable because it gives a standardized way to score explanation alignment and faithfulness rather than relying only on visual inspection. [\[9\]](#) [\[3\]](#) [\[10\]](#) [\[12\]](#) [\[13\]](#) [\[5\]](#)

Useful benchmarks and code

Saliency-Bench is the most directly useful benchmark-style resource for explanation evaluation because it combines annotated datasets with standardized faithfulness/alignment metrics. TAM’s GitHub implementation is relevant because it provides practical code for generating token activation maps and visualizing intermediate records across generation rounds. For grounding-specific evaluation, Qwen2.5-VL’s structured coordinate grounding and VPP-LLaVA-style position prompts are useful patterns for creating prompt-ablation protocols around localization. [\[8\]](#) [\[10\]](#) [\[12\]](#) [\[13\]](#) [\[6\]](#)

Open gaps

The biggest gap is that prompt studies in MLLMs usually optimize accuracy or grounding, but do not systematically test whether explanations remain stable under paraphrase, decomposition, ordering changes, role prompting, or self-refinement. A second gap is that many saliency/attribution methods are evaluated on single prompts or single outputs, not on **prompt families** that isolate one intervention at a time. A third gap is the lack of a shared benchmark for “explanation invariance under prompt intervention,” especially for TAM-like token-level maps where one would want agreement across prompts, rounds, and decoding settings. [\[3\]](#) [\[12\]](#) [\[13\]](#) [\[2\]](#) [\[1\]](#)

Most promising research direction

For stress-testing explainability in MLLMs, the most compelling design is a prompt-family benchmark where you hold the image fixed and systematically vary wording, order, decomposition, role, and CoT constraints, then score three things: answer quality, grounding accuracy, and explanation stability. The current literature suggests that performance can improve even when internal explanations remain biased, fragile, or highly prompt-dependent. That makes your subtopic well-motivated: prompt ablation is not only a performance question, but also a test of whether multimodal explanations are faithful or merely plausible. [\[2\]](#) [\[5\]](#) [\[4\]](#) [\[3\]](#)

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1. <https://arxiv.org/abs/2409.15310>
2. <https://www.arxiv.org/abs/2412.02104>
3. <https://arxiv.org/abs/2506.23270>
4. <https://arxiv.org/html/2407.04681v1>
5. <https://arxiv.org/html/2312.11570v1>
6. <https://qwenlm.github.io/blog/qwen2.5-vl/>
7. <https://arxiv.org/html/2409.12191v1>
8. <https://arxiv.org/html/2503.15426v3>
9. <https://arxiv.org/html/2506.18985v2>
10. <https://github.com/xmed-lab/TAM>
11. <https://arxiv.org/html/2602.20878v1>
12. <https://arxiv.org/html/2310.08537v3>
13. <https://arxiv.org/html/2310.08537>
14. <https://arxiv.org/html/2602.20853v1>
15. <https://www.arxiv.org/pdf/2508.20279.pdf>
16. https://mn.cs.tsinghua.edu.cn/xinwang/PDF/papers/2025_Localizing_Step-by-Step_Multimodal_Long_Video_Temporal_Grounding_with_LLM.pdf
17. <https://arxiv.org/html/2510.09135v1>
18. <https://aclanthology.org/2024.findings-acl.807.pdf>
19. <https://github.com/QwenLM/Qwen3-VL/issues/721>
20. <https://github.com/waynetomas/vpp-llava>
21. <https://github.com/QwenLM/Qwen3-VL/issues/837>
22. <https://arxiv.org/html/2403.00219v1>
23. <https://www.ijcai.org/proceedings/2025/0853.pdf>
24. <https://learnopencv.com/object-detection-with-vlms-ft-qwen2-5-vl/>
25. https://openaccess.thecvf.com/content/ICCV2025/papers/Singha_FedMVP_Federated_Multimodal_Visual_Prompt_Tuning_for_Vision-Language_Models_ICCV_2025_paper.pdf
26. <https://bohrium.dp.tech/paper/arxiv/2405.14612>
27. <https://www.emergentmind.com/papers/2412.02104>
28. <https://arxiv.org/html/2411.17237v4>
29. https://openaccess.thecvf.com/content/ICCV2025W/MMFM/papers/SR_Evaluating_Variance_in_Visual_Question_Answering_Benchmarks_ICCVW_2025_paper.pdf
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